NLA Note

1 Background

Floating Point ($\mathbb{F}$): A number = mantissa $\times$ base$^{\text{exponent}}$ (e.g. $2048 = 1 \times 2^{11}$ mantissa=1, base=2, exponent=11)

Assumptions: There is $\varepsilon_m > 0$ **(machine epsilon)** $\varepsilon_m \approx 2.22 \times 10^{-16}$

A1: $\forall \alpha \in \mathbb{R}, \exists \varepsilon \in (-\varepsilon_m, \varepsilon_m)$ s.t. $fl : \mathbb{R} \to \mathbb{F}$ by $fl(\alpha) = \alpha(1+\varepsilon)$

A2: $\forall \alpha, \beta \in \mathbb{F}, \forall * \in \{+, -, \times, \div\}, \exists \varepsilon \in (-\varepsilon_m, \varepsilon_m)$ s.t. $\alpha \circledast \beta = (\alpha * \beta)(1+\varepsilon)$

A1: The rounding error in α is relative to the size of α ($|\alpha - fl(\alpha)| < |\alpha|\varepsilon_m$) A2: The error in $\alpha \circledast \beta$ is relative to the size of $\alpha * \beta$ ($|\alpha \circledast \beta - \alpha * \beta| \le |\alpha * \beta|\varepsilon_m$)

Cost of Algorithm (C): $C :=$ number of floating point operations (FLOPs) If $C(n) \le \alpha f(n), \Rightarrow C(n) = \mathcal{O}(f(n))$.

We consider that: 交换/赋值/相反数 no cost (i.e. Permutation $\mathbf{P}$ is free), 比较两数 1 FLOPs, $\sqrt{\ }$ 1 FLOPs.

Norms: $\|\mathbf{x}\|_p := (\sum_{i=1}^n |x_i|^p)^{1/p}, p \ge 1$ $\|\mathbf{x}\|_\infty := \max_{1\le i\le n} |x_i|$ $\|\mathbf{x}\|_2 := (\sum_{i=1}^n |x_i|^2)^{1/2}$ $\|\mathbf{x}\|_1 := \sum_{i=1}^n |x_i|$

1. $\|\mathbf{x}\|_p \ge 0$ $\|\mathbf{x}\|_p = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$
2. $\|\alpha\mathbf{x}\|_p = |\alpha|\|\mathbf{x}\|_p$ for $\alpha \in \mathbb{R}$
3. $\|\mathbf{x}+\mathbf{y}\|_p \le \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$ (Triangle inequality)
4. **Induced Norm:** $\|\mathbf{A}\|_p := \sup_{\mathbf{x}\neq 0} \frac{\|\mathbf{Ax}\|_p}{\|\mathbf{x}\|_p} = \sup_{\|\mathbf{x}\|_p=1} \|\mathbf{Ax}\|_p$

1. $\|\mathbf{A}_{m\times n}\|_\infty := \max\limits_{1\le j\le m} \sum_{i=1}^n |a_{ij}|$ (maximum row sum)
2. $\|\mathbf{A}_{m\times n}\|_1 := \max\limits_{1\le i\le n} \sum_{j=1}^m |a_{ij}|$ (maximum column sum)
3. $\|\mathbf{A}_{m\times n}\|_2 := \sqrt{\rho(\mathbf{A}^\top\mathbf{A})}$ $\rho(\mathbf{A}^\top\mathbf{A}) = \max\{|\lambda| : \lambda \text{ is eigenvalue of } \mathbf{A}^\top\mathbf{A}\}$

For Matrix: $\|\mathbf{Ax}\|_p \le \|\mathbf{A}\|_p\|\mathbf{x}\|_p$ $\|\mathbf{AB}\|_p \le \|\mathbf{A}\|_p\|\mathbf{B}\|_p$ $\|\mathbf{A}^k\|_p \le \|\mathbf{A}\|_p^k$ $\|\mathbf{A}+\mathbf{B}\|_p \le \|\mathbf{A}\|_p + \|\mathbf{B}\|_p$

2 Methods for Systems of Linear Equations (SLE)

Ideal: We want to solve SLE: $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}_{n\times n}$ is invertible, $\mathbf{x}, \mathbf{b} \in \mathbb{R}^n$

2.1 Backward Stable | Condition Number | Error

Backward Stable: Algorithm solve SLE is Backward Stable if: $\exists \alpha(n) > 0$ s.t. $\forall \mathbf{A}, \mathbf{b}$, computed solution $\hat{\mathbf{x}}$ has: $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b} + \Delta\mathbf{b}$ and $\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p} + \frac{\|\Delta\mathbf{b}\|_p}{\|\mathbf{b}\|_p} \le \alpha\varepsilon_m$ ε_m : machine epsilon.

Condition Number ($\kappa_p(\mathbf{A})$): $\kappa_p(\mathbf{A}) := \begin{cases} \|\mathbf{A}\|_p\|\mathbf{A}^{-1}\|_p \ge 1 & \text{if } \mathbf{A} \text{ is invertible} \\ \infty & \text{otherwise} \end{cases}$ $\begin{cases} \text{If } \kappa_p(\mathbf{A}) \text{ is large (e.g. } \kappa_p(\mathbf{A}) \ge 10^{12} \approx \varepsilon_m^{-1}), & \mathbf{A} \text{ is ill-conditioned.} \\ \text{If } \kappa_p(\mathbf{A}) \le 10^4, & \mathbf{A} \text{ is well-conditioned.} \end{cases}$ ps: $\kappa_p(\mathbf{A})$ 衡量矩阵奇异性

If $\mathbf{A}_{n\times n}$ s.t. $|\lambda_1| \ge |\lambda_2| \ge \cdots \ge |\lambda_n|$ corresponding $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are orthonormal set. **(e.g. $\overline{A}^\top = A$)** Then $\kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2\|\mathbf{A}^{-1}\|_2 = \frac{|\lambda_1|}{|\lambda_n|} \ge 1$

Relative Error: $\frac{\|\hat{\mathbf{x}}-\mathbf{x}\|_p}{\|\mathbf{x}\|_p}$ **Actual Error:** $\|\hat{\mathbf{x}}-\mathbf{x}\|_p$ Let $\mathbf{Ax} = \mathbf{b}$ and $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$. If $\kappa_p(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p} < 1$, then: $\frac{\|\mathbf{x}-\hat{\mathbf{x}}\|_p}{\|\mathbf{x}\|_p} \le \frac{\kappa_p(\mathbf{A})}{1-\kappa_p(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p}} \cdot \frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p}$

Generally, Let $\mathbf{Ax} = \mathbf{b}$ and $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b} + \Delta\mathbf{b}$. If $\kappa_p(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p} < 1$ then $\frac{\|\mathbf{x}-\hat{\mathbf{x}}\|_p}{\|\mathbf{x}\|_p} \le \frac{\kappa_p(\mathbf{A})}{1-\kappa_p(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p}} \cdot \left(\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p} + \frac{\|\Delta\mathbf{b}\|_p}{\|\mathbf{b}\|_p}\right)$ ps: $\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p}$ $\frac{\|\Delta\mathbf{b}\|_p}{\|\mathbf{b}\|_p}$ 小代表 good stability.

2.2 LU Factorization | Gaussian Elimination (GE)

Def of LU: If $\mathbf{A}_{n\times n}$ is invertible, then $^1\exists\, \mathbf{L}$: 单位下三角, $^2\exists\, \mathbf{U}$: 可逆上三角. s.t. $\mathbf{A} = \mathbf{LU}$ **(unique)**

LU Factorization: By Gaussian elimination, $\mathbf{A}_{n\times n} = (\prod_{k=1}^{n-1} \mathbf{L}_k)^{-1}\mathbf{U} =: \mathbf{LU}$ $\mathbf{L}_k$: Lower triangular with 1s on the diagonal (by row operation on single column)

1. 如果 $\mathbf{L}$ 是单位下三角矩阵, 且只有第 k 列有非零元素在对角线下, 则 $\mathbf{L}^{-1}$ 矩阵是 $\mathbf{L}$ 对角线不动, 其余为相反数.
2. 如果 $\mathbf{L}_1, \mathbf{L}_2$ 矩阵是单位下三角矩阵, 且 $\mathbf{L}_1|\mathbf{L}_2$ 仅在第 $1 \sim k|k+1 \sim n$ 列有非零元素在对角线下, 则 $\mathbf{L}_1\mathbf{L}_2$ 矩阵是他们的" 拼接".

Error Analysis of GE: Gaussian elimination is an **unstable** algorithm. (i.e. floating point arithmetic can have a disastrous effect on the computed solution.)

2.3 LUPP | LUCP | GEPP | GECP

Permutation Matrix: For a permutation π, $\mathbf{P}_\pi := [\mathbf{e}_{\pi(1)}, \dots, \mathbf{e}_{\pi(n)}]$ ps: 写的时候是对 I, 第 i 行的 1 移动到第 $\pi(i)$ 行. 注意: 用另一方式定义的 $\mathbf{P}_\pi$ 会让后面对 π 结论不一样

· 左边乘 $\mathbf{P}_\pi$: 交换行 ($\mathbf{P}_\pi$ 的行 代表具体行的交换方法, π 也代表交换行的方法) · 右边乘 $\mathbf{P}_\pi$: 交换列 ($\mathbf{P}_\pi$ 的列 代表具体列的交换方法, π^{-1} 才代表交换列的方法)

· Proposition: $\mathbf{P}_\pi$ is orthogonal, i.e. $\mathbf{P}_\pi^\top = \mathbf{P}_\pi^{-1}$ · 对于二阶置换矩阵 **(Involution)**: $\mathbf{P}_\pi^2 = \mathbf{I}$, i.e. $\mathbf{P}_\pi^{-1} = \mathbf{P}_\pi$ · **GEPP|GECP**: 交换行 (行和列) 使每次 $|u_{kk}|$ 最大**(stable)**

Backward Stable of GEPP: $\forall$ SLE sol $\hat{\mathbf{x}}$ by GEPP, $\exists \Delta\mathbf{A}$ s.t. $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$ and $\frac{\|\Delta\mathbf{A}\|_\infty}{\|\mathbf{A}\|_\infty} \le p(n)2^{n-1}\varepsilon_m$ $p(n)$ is a cubic polynomial

2.4 QR Factorization Method

QR Factorization: QR factorization of $\mathbf{A}_{n\times n}$ is $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q}$ is orthogonal (i.e. $\mathbf{Q}^T = \mathbf{Q}^{-1}$) and $\mathbf{R}$ is non-singular upper triangular.

· Def of $\mathbf{Q}$ is orthogonal is equivalent to $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$. (i.e. columns of $\mathbf{Q}$ forming an orthonormal set.) · QR produce **large errors**

· **Orthogonal Projection:** $\text{proj}_\mathbf{u}(\mathbf{x}) = \frac{\mathbf{x}\cdot\mathbf{u}}{\|\mathbf{u}\|_2^2}\mathbf{u} = (\mathbf{u}^\top\mathbf{x})\mathbf{u}$ if $\|\mathbf{u}\|_2 = 1$. $\mathbf{x} = \text{proj}_\mathbf{u}(\mathbf{x})^{\text{parallel to } \mathbf{u}} + (\mathbf{x} - \text{proj}_\mathbf{u}(\mathbf{x}))^{\text{orthogonal to } \mathbf{u}}$

Error Analysis of MGS: The matrices $\widehat{\mathbf{Q}}$ and $\widehat{\mathbf{Q}}$ computed by MGS satisfies: $\widehat{\mathbf{Q}}^\top\widehat{\mathbf{Q}} = \mathbf{I} + \Delta\mathbf{I}$ $\widehat{\mathbf{Q}}\widehat{\mathbf{R}} = \mathbf{A} + \Delta\mathbf{A}$ where:

$\|\Delta\mathbf{I}\|_2 \le \frac{\alpha_1(n)\varepsilon_m\kappa_2(\mathbf{A})}{1-\alpha_1(n)\varepsilon_m\kappa_2(\mathbf{A})}$, if $1 - \alpha_1(n)\varepsilon_m\kappa_2(\mathbf{A}) > 0$ $\|\Delta\mathbf{A}\|_2 \le \alpha_2(n)\varepsilon_m\|\mathbf{A}\|_2$ ps: $\alpha_1(n), \alpha_2(n)$ 只由 n 决定

2.5 Householder QR factorisation

Householder QR: (Most) **stable** algorithm **Householder Reflection Matrix**: $\mathbf{H}(\mathbf{w}) := \mathbf{I} - 2\frac{\mathbf{ww}^\top}{\|\mathbf{w}\|_2^2}$ $\mathbf{H}(\mathbf{w})\mathbf{x} = \mathbf{x} - 2\text{proj}_\mathbf{w}(\mathbf{x})$

· $\mathbf{H}(\mathbf{w})$ is **orthogonal** and **symmetric** (In the Algorithm is $\mathbf{Q}_k, \mathbf{H}_k$) · $\mathbf{H}(\mathbf{w})\mathbf{x}$ 作用为让 $\mathbf{x}$ 向量对$\mathbf{w}$ 的法向量 (超平面) 上进行反射

Error Analysis of Householder QR: Let $\hat{\mathbf{x}}$ denote sol of $\mathbf{Ax} = \mathbf{b}$ by Householder QR+SQR. Then $(\mathbf{A}+\Delta\mathbf{A}) = \mathbf{b}$, where $\|\Delta\mathbf{A}\|_2 \le \alpha n^{\frac{5}{2}}\varepsilon_m\|\mathbf{A}\|_2$ α small constant

2.6 Iterative Methods for SLE

Iterative Method: For SLE $\mathbf{Ax} = \mathbf{b}$, if $\mathbf{A} = \mathbf{M} + \mathbf{N}$ with $\mathbf{M}$ is *simpler structure* than $\mathbf{A}$. (例如: 对角/三角/对称矩阵) $\Rightarrow$ 迭代求解近似解 Diff choices $\mathbf{M}, \mathbf{N} \to$ different iterative methods.

Error Analysis: Error $\mathbf{e}_k := \mathbf{x} - \mathbf{x}_k \Rightarrow \mathbf{e}_k = \mathbf{R}^k\mathbf{e}_0$ $\mathbf{R} := -\mathbf{M}^{-1}\mathbf{N}$ **Theorem**: If $\|\mathbf{R}\|_p < 1$, then $\lim\limits_{k\to\infty} \|\mathbf{e}_k\|_p = 0 \Leftrightarrow \lim\limits_{k\to\infty} \mathbf{x}_k = \mathbf{x}$ (Convergence) **GAU 收敛快于 JAC**

JAC|GAU: If $\mathbf{A}$ is **strictly diagonally dominant** [i.e. $|a_{ii}| > \sum_{j\neq i}|a_{ij}|$ $\forall i$ (对角线元素大于该行其他元素之和)], then converges for $p = \infty$. For $p = 1$ 看列和/转置.

3 Polynomial Fitting & LSQ

ps: 范德蒙德矩阵 **ill-cond**

Vandermonde Matrix: $A \in \mathbb{R}^{n\times m}$ s.t. $a_{ij} = a_i^{j-1}$ **Poly:** Deg $m-1$ poly fit to points: $\{(a_i, b_i)\}_{i=1}^n$ by solving $\mathbf{Ax} = \mathbf{b} \Rightarrow$ sol $\mathbf{x}$ is coeff. of poly $p(x) = \sum_{j=1}^m x_j x^{j-1}$

LSQ: Finding *line* which best fits $\{(a_i, b_i)\}_{i=1}^n$ by *minimise* $\|\mathbf{Ax}-\mathbf{b}\|_2$, $A \in \mathbb{R}^{n\times 2}$ vandermonde. **Thm**: *minimises* $\|\mathbf{Ax}-\mathbf{b}\|_2 \Leftrightarrow$ solve **Normal Equations** $\mathbf{A}^\top\mathbf{Ax} = \mathbf{A}^\top\mathbf{b}$

Moore-Penrose Pseudo-inverse of A: $\mathbf{A}^\dagger = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$ · *Normal Equations* has **unique sol** if $\mathbf{AA}^\top$ invertible. · If $\mathbf{A}$ invertible, $\mathbf{A}^\dagger = \mathbf{A}^{-1}$

3.1 Singular Value Decomposition (SVD)

New Def of Condition Number: If $\mathbf{A}_{m\times n}, m \ge n$ s.t. it's full rank **(i.e. rank(A) = n)**, then $\kappa_p(\mathbf{A}) := \|\mathbf{A}\|_p\|\mathbf{A}^\dagger\|_p \ge 1$ $\kappa_2(\mathbf{A}^\top\mathbf{A}) = \kappa_2(\mathbf{A})^2$ $\kappa_2(\mathbf{A}) = \frac{\sigma_{\max}}{\sigma_{\min}}$

SVD: $\mathbf{A}_{m\times n} = \mathbf{U}\Sigma\mathbf{V}^\top$ where $\mathbf{U}_{m\times m}, \mathbf{V}_{n\times n}$ *orthogonal*, $\Sigma_{m\times n}$ diagonal with singular values $\sigma_1, \dots, \sigma_p, p = \min\{m, n\}$ $\mathbf{U}/\mathbf{V}$ 的列为 left/right singular vectors.

[1] Always exist [2] $\mathbf{Av}_i = \sigma_i\mathbf{u}_i$ and $\mathbf{A}^\top\mathbf{u}_i = \sigma_i\mathbf{v}_i$ [3] $\sigma_i^2|\mathbf{v}_i$ **are eigenvalues/vectors of** $\mathbf{A}^\top\mathbf{A}$ (i.e. $\mathbf{A}^\top\mathbf{Av}_i = \sigma_i^2\mathbf{v}_i$) [4] $\text{rank}(\mathbf{A}) = \text{rank}(\Sigma) = \#$ non-zero σ_i [5] $\mathbf{u}_i = \frac{1}{\sigma_i}\mathbf{Av}_i$

Reduced QR Factorisation: By QR: $\mathbf{A}_{m\times n} = \mathbf{Q}_{m\times m}\mathbf{R}_{m\times n}, m \ge n$. [1] R 的 n 行以下全 0 [2] $\mathbf{Q} = [\mathbf{Q}_1\ \mathbf{Q}_2], \mathbf{R} = [\mathbf{R}_1\ \mathbf{0}]^\top \Rightarrow \mathbf{A} = \mathbf{Q}_1\mathbf{R}_1$ where $\mathbf{Q}_1, \mathbf{R}_1 \in \mathbb{R}^{n\times n}$

Reduced SVD: By SVD: $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top, m \ge n$. [1] If $\mathbf{B}$ orthogonal, then $\mathbf{B}^{-1} = \mathbf{B}^\top$, $\det(\mathbf{B}) = \pm 1$ and $\|\mathbf{Bx}\|_2 = \|\mathbf{x}\|_2, \forall \mathbf{x}$ [2] $\mathbf{U} = [\mathbf{U}_1\ \mathbf{U}_2], \Sigma = [\Sigma_1\ \mathbf{0}]^\top \Rightarrow \mathbf{A} = \mathbf{U}_1\Sigma_1\mathbf{V}^\top, \mathbf{u}_1 \in \mathbb{R}^{m\times n}, \Sigma_1 \in \mathbb{R}^{n\times n}$

Error Analysis of LSQ-QR: Let $\hat{\mathbf{x}}$ denote sol of LSQ-QR (Householder). Then $\hat{\mathbf{x}}$ minimises $\|(\mathbf{A}+\Delta\mathbf{A})\mathbf{x} - \mathbf{b}\|_2$, and $\frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2} \le \alpha n^{\frac{3}{2}}m\varepsilon_m, \alpha > 0$

Error Analysis LSQ (any): Let $\mathbf{x}|\hat{\mathbf{x}}$ minimises $\|\mathbf{Ax}-\mathbf{b}\|_2|\|(\mathbf{A}+\Delta\mathbf{A})\mathbf{x}-\mathbf{b}\|_2$ and $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}+\Delta\mathbf{A}) = n$. If $\kappa_2(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2} < 1 \Rightarrow \frac{\|\mathbf{x}-\hat{\mathbf{x}}\|_2}{\|\mathbf{x}\|_2} \le \frac{\kappa_2(\mathbf{A})}{1-\kappa_2(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2}} \cdot \frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2}\left(2 + (\kappa_2(\mathbf{A})+1)\frac{\|\mathbf{r}\|_2}{\|\mathbf{A}\|_2\|\mathbf{x}\|_2}\right), \mathbf{r} = \mathbf{b} - \mathbf{Ax}$

4 Eigenvalue Problems

(In this section: $|\lambda_1| \ge |\lambda_2| \ge \cdots \ge |\lambda_n|$ are eigenvalues of $\mathbf{A} \in \mathbb{R}^{n\times n}$ with corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$. $\lambda_1 = \lambda_{\max} \leftrightarrow \mathbf{x}_{\max}, \lambda_n = \lambda_{\min} \leftrightarrow \mathbf{x}_{\min}$)

Rayleigh Quotient: $r_\mathbf{A}(\mathbf{x}) = \frac{\mathbf{x}^\top\mathbf{Ax}}{\mathbf{x}^\top\mathbf{x}}$ $r_\mathbf{A}(\mathbf{x}_i) = \lambda_i$ Eigenvalues $\leftrightarrow$ Eigenvectors: $\mathbf{x}_i \to \lambda_i$: $r_\mathbf{A}(\mathbf{x}_i) = \lambda_i$ $\lambda_i \to \mathbf{x}_i$: $(\mathbf{A} - \lambda_i\mathbf{I})\mathbf{x}_i = \mathbf{0}$ **Basis**: $\mathbf{Ax} = \sum_i \alpha_i\lambda_i\mathbf{x}_i$

Convergence of Power Iteration: If $|\lambda_1| > |\lambda_2|$ and $\mathbf{x}_1 \cdot \mathbf{z}^{(0)} \neq 0$, then $\exists \sigma_{k\,k\in\mathbb{N}}$ with $\sigma_k = \pm 1$ s.t. $\lim\limits_{k\to\infty} \|\sigma_k\mathbf{z}^{(k)} - \mathbf{x}_1\|_2 = 0$ and $\lim\limits_{k\to\infty} |\lambda^{(k)} - \lambda_1| = 0$

Useful Tool: Eigenvalues of $\mathbf{A}^{-1}$ are $\mu_i = \frac{1}{\lambda_i}$ and $\lambda_{\max} = \mu_{\min}^{-1}$ Eigenvalues of $(\mathbf{A} - s\mathbf{I})^{-1}$ are $\nu_i = \frac{1}{\lambda_i - s}$ and $\lambda_{\max} = s + \nu_{\min}^{-1}$

Inverse Iteration Algorithm: Input $\mathbf{A}^{-1}$ in Power Iteration $\Rightarrow$ find $\lambda_{\min}^{-1}$ and $\mathbf{x}_{\min}$ (用 SII $s = 0$ 可直接得到)

Shifted Inverse Iteration Algorithm: Using $(\mathbf{A} - s\mathbf{I})^{-1} \Rightarrow$ Find eigenvalue closest to s and corresponding eigenvector. Assume: $|\lambda_a - s| < |\lambda_b - s| < \cdots$

Rate of Convergence for PI & SII: **Rate** $\uparrow$ **iff** $|\frac{\lambda_2}{\lambda_1}| \downarrow \|\sigma_k\mathbf{z}^{(k)} - \mathbf{x}_1\|_2 \le \alpha_1(\frac{|\lambda_2|}{|\lambda_1|})^k$; $|\lambda^{(k)} - \lambda_1| \le \alpha_2(\frac{|\lambda_2|}{|\lambda_1|})^{2k}$ **Rate** $\uparrow$ **iff** $|\frac{\lambda_a - s}{\lambda_b - s}| \downarrow \|\sigma_k\mathbf{z}^{(k)} - \mathbf{x}_i\|_2 \le \beta_1(\frac{|\lambda_a - s|}{|\lambda_b - s|})^k$; $|\lambda^{(k)} - \lambda_a| \le \beta_2(\frac{|\lambda_a - s|}{|\lambda_b - s|})^{2k}$

DP Dot product $C = \sum_{i=1}^{n} 2 = 2n = \mathcal{O}(n)$

Input: $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$
Output: $s = \mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y}$

$s = 0$
for $i = 1, \dots, n$ **do**
  $s = s + x_i y_i$
end for

Rank Theorem: [$\mathbf{Ax} = \mathbf{0}$ iff $\mathbf{x} = \mathbf{0}$] $\Leftrightarrow$ [列线性无关] $\Leftrightarrow$ [$\mathbf{A}$ 可逆] $\Leftrightarrow$ [rank($\mathbf{A}$) = rank($\mathbf{A}^\top$) = n] Abel: ≥ 5 次方 | 实系数多项式 |∃ 实根无法用有限次加减乘除和开方表示.
Spectral Theorem: If $\mathbf{A}$ is symmetric, then ∃ orthonormal $\mathbf{Q}$ s.t. $\mathbf{A} = \mathbf{Q}\Lambda\mathbf{Q}^\top$ (Orthogonally diagonalisable) $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$

LU LU factorisation
$C(n) = \sum_{k=1}^{n-1}(n-k)(1+2(n-k+1)) = \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{7}{6}n$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$
Output: $\mathbf{L}, \mathbf{U} \in \mathbb{R}^{n \times n}$ s.t. $\mathbf{A} = \mathbf{LU}$

$\mathbf{L} = \mathbf{I}$, $\mathbf{U} = \mathbf{A}$
for $k = 1, \dots, n-1$ **do**
  for $j = k+1, \dots, n$ **do**
    $l_{jk} = u_{jk}/u_{kk}$
    $(u_{jk}, \dots, u_{jn}) = (u_{jk}, \dots, u_{jn}) - l_{jk}(u_{kk}, \dots, u_{kn})$
  end for
end for

FS Forward substitution
$C(n) = \sum_{j=1}^{n}[2(j-1)+1] = n^2 = \mathcal{O}(n^2)$

Input: $\mathbf{L} \in \mathbb{R}^{n \times n}$ (lower triangular), $\mathbf{b} \in \mathbb{R}^n$
Output: $\mathbf{y} \in \mathbb{R}^n$ such that $\mathbf{Ly} = \mathbf{b}$

for $j = 1, \dots, n$ **do**
  $y_j = \frac{1}{l_{jj}}\left(b_j - \sum_{k=1}^{j-1} l_{jk} y_k\right)$
end for

BS Back substitution
$C(n) = \sum_{j=1}^{n}[2(n-j)+1] = n^2 = \mathcal{O}(n^2)$

Input: $\mathbf{U} \in \mathbb{R}^{n \times n}$ (upper triangular), $\mathbf{y} \in \mathbb{R}^n$
Output: $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{Ux} = \mathbf{y}$

for $j = n, \dots, 1$ **do**
  $x_j = \frac{1}{u_{jj}}\left(y_j - \sum_{k=j+1}^{n} u_{jk} x_k\right)$
end for

GE Gaussian elimination $C(n) = \frac{2}{3}n^3 + \frac{5}{2}n^2 - \frac{7}{6}n = \mathcal{O}(n^3)$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$
Output: $\mathbf{x} \in \mathbb{R}^n$ s.t. $\mathbf{Ax} = \mathbf{b}$

Find LU factorisation of $\mathbf{A} = \mathbf{LU}$. # **LU** Algorithm
Solve $\mathbf{Ly} = \mathbf{b}$ using forward substitution. # **FS** Algorithm
Solve $\mathbf{Ux} = \mathbf{y}$ using back substitution. # **BS** Algorithm

GEPP GE(partial pivoting) $C(n) = \frac{2}{3}n^3 + 3n^2 - \frac{5}{3}n = \mathcal{O}(n^3)$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$ # numpy.linalg.solve use it
Output: $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{Ax} = \mathbf{b}$

Find $\mathbf{PA} = \mathbf{LU}$ using Algorithm LUPP # **LUPP** Algorithm
Solve $\mathbf{Ly} = \mathbf{Pb}$ by forward substitution # **FS** Algorithm
Solve $\mathbf{Ux} = \mathbf{y}$ by back substitution # **BS** Algorithm

GECP GE(complete pivoting) $C(n) = n^3 + 3n^2 - 2n = \mathcal{O}(n^3)$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$
Output: $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{Ax} = \mathbf{b}$

Find $\mathbf{PAQ} = \mathbf{LU}$ using Algorithm LUCP # **LUCP** Algorithm
Solve $\mathbf{Ly} = \mathbf{Pb}$ by forward substitution # **FS** Algorithm
Solve $\mathbf{Uz} = \mathbf{y}$ by back substitution # **BS** Algorithm
Compute $\mathbf{x} = \mathbf{Qz}$

LUPP LU factorisation with partial pivoting (not unique) $C(n) = \frac{2}{3}n^3 + n^2 - \frac{5}{3}n$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$
Output: $\mathbf{L}$, $\mathbf{U}$, $\mathbf{P} \in \mathbb{R}^{n \times n}$ s.t. $\mathbf{PA} = \mathbf{LU}$ with $\mathbf{L}$ 单位下三角, $\mathbf{U}$ 可逆上三角

$\mathbf{U} = \mathbf{A}$, $\mathbf{L} = \mathbf{I}$, $\mathbf{P} = \mathbf{I}$
for $k = 1, \dots, n-1$ **do**
  Choose $i \in \{k, \dots, n\}$ which maximizes $|u_{ik}|$ # **Max** Algorithm
  Exchange row $(u_{kk}, \dots, u_{kn})$ with $(u_{ik}, \dots, u_{in})$ # 第 k 行和第 i 行 的 $k \sim n$ 列交换
  Exchange row $(l_{k1}, \dots, l_{k,k-1})$ with $(l_{i1}, \dots, l_{i,k-1})$ # 第 k 行和第 i 行 的 $1 \sim k-1$ 列交换
  Exchange row $(p_{k1}, \dots, p_{kn})$ with $(p_{i1}, \dots, p_{in})$ # 第 k 行和第 i 行 的全部列交换
  for $j = k+1, \dots, n$ **do**
    $l_{jk} = u_{jk}/u_{kk}$
    $(u_{jk}, \dots, u_{jn}) = (u_{jk}, \dots, u_{jn}) - l_{jk}(u_{kk}, \dots, u_{kn})$
  end for
end for

LUCP LU factorisation with complete pivoting (not unique) $C(n) = n^3 + n^2 - 2n$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$
Output: $\mathbf{L}$, $\mathbf{U}$, $\mathbf{P}$, $\mathbf{Q} \in \mathbb{R}^{n \times n}$ s.t. $\mathbf{PAQ} = \mathbf{LU}$ with $\mathbf{L}$ 单位下三角, $\mathbf{U}$ 可逆上三角

$\mathbf{U} = \mathbf{A}$, $\mathbf{L} = \mathbf{I}$, $\mathbf{P} = \mathbf{I}$, $\mathbf{Q} = \mathbf{I}$
for $k = 1, \dots, n-1$ **do**
  Choose $(i, j) \in \{(k, k), \dots, (n, n)\}$ which maximizes $|u_{ij}|$ # **Max** Algorithm
  Exchange row $(u_{kk}, \dots, u_{kn})$ with $(u_{ik}, \dots, u_{in})$ # 第 k 行和第 i 行 的 $k \sim n$ 列交换 U
  Exchange row $(l_{k1}, \dots, l_{k,k-1})$ with $(l_{i1}, \dots, l_{i,k-1})$ # 第 k 行和第 i 行 的 $1 \sim k-1$ 列交换 L
  Exchange row $(p_{k1}, \dots, p_{kn})$ with $(p_{i1}, \dots, p_{in})$ # 第 k 行和第 i 行 的全部列交换 P
  Exchange column $(u_{1k}, \dots, u_{nk})$ with $(u_{1j}, \dots, u_{nj})$ # 第 k 列和第 j 列 的全部行交换 U
  Exchange column $(q_{1k}, \dots, q_{nk})$ with $(q_{1j}, \dots, q_{nj})$ # 第 k 列和第 j 列 的全部行交换 Q
  for $m = k+1, \dots, n$ **do**
    $l_{mk} = u_{mk}/u_{kk}$
    $(u_{mk}, \dots, u_{mn}) = (u_{mk}, \dots, u_{mn}) - l_{mk}(u_{kk}, \dots, u_{kn})$
  end for
end for

(M)GS SQR Householder-QR

SQR (SLE) via QR-Factorization $C(n) = 2n^3 + 4n^2 + n$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$
Output: $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{Ax} = \mathbf{b}$

Find QR factorization of $\mathbf{A}$ # **GS**|**MGS** Algorithm
Solve $\mathbf{Qy} = \mathbf{b}$ # **MV** Algorithm # ps: $\mathbf{y} = \mathbf{Q}^{-1}\mathbf{b} = \mathbf{Q}^\top\mathbf{b}$
Solve $\mathbf{Rx} = \mathbf{y}$ using Algorithm BS # **BS** Algorithm

Sum Notation: $\sum_{k=1}^{n} k = \frac{n(n+1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n$
$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6} = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n$
$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2 = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2$
Rank Theorem: $\mathbf{A}_{m \times n}$, $m \geq n$: [$\mathbf{Ax} = \mathbf{0}$ iff $\mathbf{x} = \mathbf{0}$] $\Leftrightarrow$ [rank($\mathbf{A}$) = n]
Cauchy-Schwarz Inequality: For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $|\mathbf{x} \cdot \mathbf{y}| = |\mathbf{x}^\top \mathbf{y}| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$
Determinant: $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ Orthogonal: If $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}$
Eigen: $\mathbf{Av} = \lambda\mathbf{v}$ $\xi_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = \prod_i(\lambda_i - \lambda)$

Iterative Method | OP MAX

GI|JAC|GAU JAC: $C = \mathcal{O}(k_{\max} n^2)$ | GAU: $C = \mathcal{O}(k_{\max} n^2)$

Input: $\mathbf{A} = \mathbf{M} + \mathbf{N} \in \mathbb{R}^{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$, $\mathbf{x}_0 \in \mathbb{R}^n$, $\varepsilon_r > 0$
Output: $\mathbf{x}_k \in \mathbb{R}^n$ with $\mathbf{Ax}_k \approx \mathbf{b}$

$k = 0$ # JAC: $C = 2n^2$/it GAU: $C = 2n^2$/it
while $\|\mathbf{Ax}_k - \mathbf{b}\|_p > \varepsilon_r \|\mathbf{b}\|_p$ **do** # e.g. $\varepsilon_r = 10^{-3}$
  $k = k + 1$ # 上面给的 C 不含这步
  Compute $\mathbf{y}_k = \mathbf{b} - \mathbf{N}\mathbf{x}_{k-1}$
  Solve $\mathbf{M}\mathbf{x}_k = \mathbf{y}_k$ # $\downarrow \mathbf{A} = \mathbf{D} + \mathbf{L} + \mathbf{U}$
end while # JAC: $M = D$, $N = L + U$; GAU: $M = D + L$, $N = U$

LSQ-QR LSQ via QR $C = 2mn^2 - \frac{2}{3}n^3 + \mathcal{O}(m^2 + n^2 + mn)$

Input: $\mathbf{A} \in \mathbb{R}^{m \times n}$, $m \geq n$, rank($\mathbf{A}$) = n, $\mathbf{b} \in \mathbb{R}^m$ ↑ by reduced QR
Output: $\mathbf{x} \in \mathbb{R}^n$ with $\|\mathbf{Ax} - \mathbf{b}\|_2$ minimised

Compute reduced QR factorization $\mathbf{A} = \mathbf{Q}_1\mathbf{R}_1$ # **QR** Algorithm
Compute $\mathbf{y} = \mathbf{Q}_1^\top\mathbf{b}$ # **MV** Algorithm # $C = 2mn$
Solve $\mathbf{R}_1\mathbf{x} = \mathbf{y}$ # **BS** Algorithm # $C = n^2$

PI Power Iteration $C = ?$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$ *symmetric*, $\mathbf{z}^{(0)} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, $\varepsilon > 0$
Output: $\mathbf{z}^{(k)} \in \mathbb{R}^n$, $\lambda^{(k)} \in \mathbb{R}$ s.t. $\mathbf{z}^{(k)} \approx \mathbf{x}_{\max}$, $\lambda^{(k)} \approx \lambda_{\max}$

$k = 0$
$\lambda^{(0)} = r_{\mathbf{A}}(\mathbf{z}^{(0)})$
while $\|\mathbf{Az}^{(k)} - \lambda^{(k)}\mathbf{z}^{(k)}\|_2 > \varepsilon$ **do** # or: $|\lambda^{(k)} - \lambda^{(k-1)}| > \varepsilon$
  $k = k + 1$ # or: $\|\mathbf{z}^{(k)} - \mathbf{z}^{(k-1)}\|_2 > \varepsilon$
  $\mathbf{z}^{(k)} = \mathbf{Az}^{(k-1)}$
  $\mathbf{z}^{(k)} = \frac{\mathbf{z}^{(k)}}{\|\mathbf{z}^{(k)}\|_2}$ # To improve stability and avoid large values
  $\lambda^{(k)} = r_{\mathbf{A}}(\mathbf{z}^{(k)})$ # In fact: $\mathbf{z}^{(k)} = \frac{\mathbf{A}^k\mathbf{z}^{(0)}}{\|\mathbf{A}^k\mathbf{z}^{(0)}\|_2}$ and $\lambda^{(k)} = r_{\mathbf{A}}(\mathbf{z}^{(k)})$
end while

(M)GS (Modified) Gram-Schmidt $C(n) = 2n^3 + n^2 + n$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$
Output: $\mathbf{Q} \in \mathbb{R}^{n \times n}$, with orthonormal columns $\mathbf{q}_1, \dots, \mathbf{q}_n$
  $\mathbf{R} \in \mathbb{R}^{n \times n}$ non-singular upper triangular matrix; s.t. $\mathbf{A} = \mathbf{QR}$

$\mathbf{R} = \mathbf{0}$
for $j = 1, \dots, n$ **do**
  $\mathbf{q}_j = \mathbf{a}_j$
  for $k = 1, \dots, j-1$ **do**
    $r_{kj} = \mathbf{q}_k^\top\mathbf{a}_j$ | $(r_{kj} = \mathbf{q}_k^\top\mathbf{q}_j)$
    $\mathbf{q}_j = \mathbf{q}_j - r_{kj}\mathbf{q}_k$
  end for
  $r_{jj} = \|\mathbf{q}_j\|_2 = \sqrt{\mathbf{q}_j^\top\mathbf{q}_j}$
  $\mathbf{q}_j = \frac{\mathbf{q}_j}{r_{jj}}$
end for

OP Outer product $C = \sum_{i=1}^{n}\sum_{j=1}^{n} 1 = n^2 = \mathcal{O}(n^2)$

Input: $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$
Output: $A = \mathbf{xy}^\top$

for $i = 1, \dots, n$ **do**
  for $j = 1, \dots, n$ **do**
    $a_{ij} = x_i y_j$
  end for
end for

LSQ-SVD LSQ via SVD

Input: $\mathbf{A} \in \mathbb{R}^{m \times n}$, $m \geq n$, rank($\mathbf{A}$) = n, $\mathbf{b} \in \mathbb{R}^m$
Output: $\mathbf{x} \in \mathbb{R}^n$ with $\|\mathbf{Ax} - \mathbf{b}\|_2$ minimised

Compute reduced SVD $\mathbf{A} = \mathbf{U}_1\Sigma_1\mathbf{V}^\top$ # see SVD note
Compute $\mathbf{z} = \mathbf{U}_1^\top\mathbf{b}$ # **MV** Algorithm # $C = 2mn$
Solve $\Sigma_1\mathbf{y} = \mathbf{z}$ # $C = n$
Compute $\mathbf{x} = \mathbf{Vy}$ # **MV** Algorithm # $C = 2n^2$

SII Shifted Inverse Iteration $C = ?$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$ *symmetric*, $s \in \mathbb{R}$, $\mathbf{z}^{(0)} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, $\varepsilon > 0$
Output: $\mathbf{z}^{(k)} \in \mathbb{R}^n$, $\lambda^{(k)} \in \mathbb{R}$ s.t. $\mathbf{z}^{(k)} \approx \mathbf{x}_a$, $\lambda^{(k)} \approx \lambda_a$ near s

$k = 0$
$\lambda^{(0)} = r_{\mathbf{A}}(\mathbf{z}^{(0)})$ # first $O(n^3)$ cost; then $O(n^2)$ cost/it
while $\|\mathbf{Az}^{(k)} - \lambda^{(k)}\mathbf{z}^{(k)}\|_2 > \varepsilon$ **do**
  $k = k + 1$
  Solve $(\mathbf{A} - s\mathbf{I})\mathbf{y} = \mathbf{z}^{(k-1)}$ for $\mathbf{y}$ # via GEPP & Find LU only once
  $\mathbf{z}^{(k)} = \frac{\mathbf{y}}{\|\mathbf{y}\|_2}$
  $\lambda^{(k)} = r_{\mathbf{A}}(\mathbf{z}^{(k)})$ # In fact: $\mathbf{z}^{(k)} = \frac{(\mathbf{A}-s\mathbf{I})^{-k}\mathbf{z}^{(0)}}{\|(\mathbf{A}-s\mathbf{I})^{-k}\mathbf{z}^{(0)}\|_2}$, $\lambda^{(k)} = r_{\mathbf{A}}(\mathbf{z}^{(k)})$
end while

QR Householder QR factorization $C(n) = \frac{8}{3}n^3 + \mathcal{O}(n^2)$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$
Output: $\mathbf{Q} \in \mathbb{R}^{n \times n}$ orthogonal
  $\mathbf{R} \in \mathbb{R}^{n \times n}$ non-singular upper triangular matrix; s.t. $\mathbf{A} = \mathbf{QR}$

$\mathbf{Q} = \mathbf{I}$, $\mathbf{R} = \mathbf{A}$ # 如果 $A \in \mathbb{R}^{m \times n}$, $m \geq n$, loop 到 n, 且 3.6.7 的 n 改为 m
for $k = 1, \dots, n-1$ **do**
  $\mathbf{u} = (r_{kk}, \dots, r_{nk}) \in \mathbb{R}^{n-k+1}$
  $\mathbf{v} = \mathbf{u} + \text{sign}(u_1)\|\mathbf{u}\|_2\mathbf{e}_1$ # line.4-7 得到的 $\mathbf{Q}_k$:
  $\mathbf{w} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}$ # 使得 $\mathbf{Q}_k\mathbf{u} = -\text{sign}(u_1)\|\mathbf{u}\|_2\mathbf{e}_1$
  $\mathbf{H}_k = \mathbf{I} - 2\mathbf{ww}^\top \in \mathbb{R}^{(n-k+1)\times(n-k+1)}$ # line.6-8 一起算, 见下
  $\mathbf{Q}_k = \mathbf{I}_{k-1} \oplus \mathbf{H}_k \in \mathbb{R}^{n \times n}$ # ↓line.6-8 $C = 4(n-k+1)^2 + (n-k+1)$
  $\mathbf{R} = \mathbf{Q}_k\mathbf{R}$ # 用矩阵分块, 只有右下角 $\mathbf{H}_k\mathbf{R}_{k:n,k:n}$ 要算 → 用 H_k 定义 Alg. MV+OP
  $\mathbf{Q} = \mathbf{QQ}_k$ # If using in Alg. **SQR**, then $9' : \mathbf{y} = \mathbf{Q}_k\mathbf{y}$; $C = \frac{4}{3}n^3 + \mathcal{O}(n^2)$ (SQR/QR)
end for # $\mathbf{A} \in \mathbb{R}^{m \times n}$ reduced 的 $C = 2n^2m - \frac{2}{3}n^3 + \mathcal{O}(m^2 + n^2 + mn)$

Max Finding index of maximum $C(m) = m - 1$

Input: $z \in \mathbb{R}^m$
Output: index i with $|z_i| = \max_{j=1,\dots,m}|z_j|$

$i = 1$, max = $|z_1|$
for $j = 2, \dots, m$ **do**
  if $|z_j| >$ max **then**
    $i = j$
    max = $|z_j|$
  end if
end for

5 Appendix

5.1 Useful Theorems/Formulas

5.2 Others

Where the error come from calucation:

1. If your calculation involves 15 or more significant digits, you will likely encounter rounding errors. (Since $\varepsilon_m \approx 10^{-16}$ (relative accuracy), which is about 15 significant digits.)
2. **For GE Algorithm**: It is an unstable algorithm. e.g. $\mathbf{A} = [10^{-20}\ 1 \setminus\setminus\ 1\ 1] \Rightarrow$ By **GE** Algorithm, $\hat{\mathbf{L}}\hat{\mathbf{U}} = \mathbf{A} + [0\ 0 \setminus\setminus\ 0\ -1]$ (error)
3. **MGS vs. GS**:In floating point arithmetic, the computed vectors $\{\hat{q}_1, \ldots, \hat{q}_{j-1}\}$ will not be orthonormal. Algorithm MGS takes this into account when computing $\hat{q}_j$; Algorithm GS does not.